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Advanced Quantitative Financial Analytics and Algorithmic Trading

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1. Overview

This report presents the backtest results for the Stochastic Oscillator Crossover strategy applied to the SPDR S&P 500 ETF Trust ("SPY"). The strategy utilizes the %K and %D lines of the Stochastic Oscillator to identify entry and exit points in a range-bound market [1]. Performance metrics and equity curves are provided, as well as the strategy's historical context and publisher insights.

2. Strategy Description

2.1 History

The **Stochastic Oscillator**, developed by **George C. Lane** in the late 1950s, is a momentum indicator that helps traders identify potential market reversals. It measures the relationship between the closing price and the price range over a specific period, typically 14 days. Lane believed that momentum changes direction before prices, making the indicator valuable for predicting market tops and bottoms. [2]

The oscillator became popular for its effectiveness in range-bound markets, identifying overbought and oversold conditions. The crossover strategy—buying when %K crosses above %D in oversold zones and selling when %K crosses below %D in overbought zones—remains widely used. The indicator is still relevant in modern technical analysis across various asset classes, including stocks, forex, and commodities. [2]

2.2 Definition

The Stochastic Oscillator is a momentum indicator that compares a specific closing price of an asset to a range of its prices over a certain period. It consists of two lines:

1. %K line: Represents the current closing price relative to the range of prices over a given period.
2. %D line: A simple moving average of the %K line, which smooths out the fluctuations for easier signal identification.

2.3 Strategy rules

Setup:

- **Overbought Zone:** 80
- **Oversold Zone:** 20
- **Lookback period (for %K)**: 14
- **Smoothing period (for %D)**: 3

Trading Signals:

Entry Rule:

• Long (Buy): When the %K line crosses above the %D line in the oversold zone (below 20).

• Short (Sell): When the %K line crosses below the %D line in the overbought zone (above 80).

Exit Rule:

• Long (Sell): When the %K line crosses below the %D line.

• Short (Buy): When the %K line crosses above the %D line

3. Backtesting

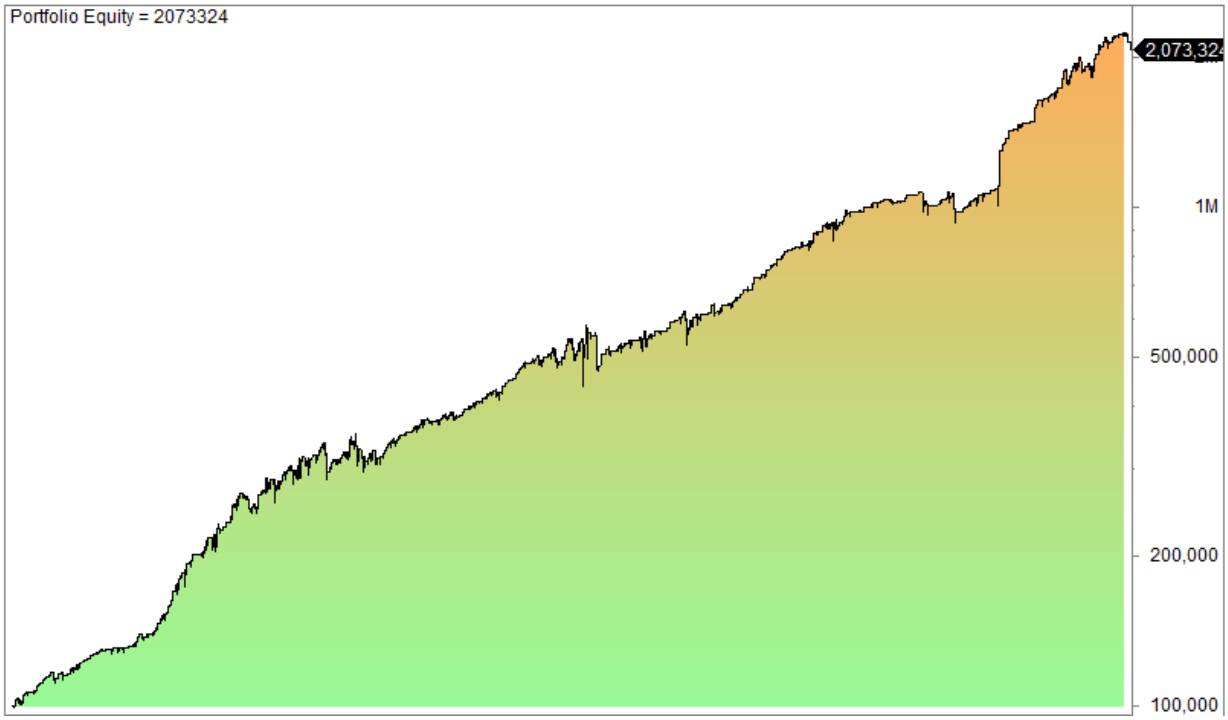
3.1. Publisher strategy Performance

According to the publisher, the Stochastic Oscillator strategy demonstrated promising performance when backtested on the S&P 500 (using SPY as a proxy) from its inception in 1993 until the time of publication (2024). [1]

Data Details

- **Asset**: SPDR S&P 500 ETF Trust ("SPY")
- **Timeframe**: Daily
- **Date Range**: January 1, 1993 - April 08, 2024
- **Backtesting Framework**: Amibroker

Equity curve



Source: [2]

Insights

The equity curve demonstrates the Stochastic Oscillator strategy's consistent long-term profitability, steady growth with manageable drawdowns, and adaptability across different market conditions, all while reflecting compound gains and stable performance.

The key performance metrics are reported in the table below:

Metric	**Values**
Number of Trades	556
Average Return per Trade	0.57%
Profit Factor	2.2
Max Drawdown	19.8%

Interpretation

- **Number of Trades**: The strategy executed a total of **556 trades** from 1993 to [end date].
- **Average Return per Trade**: The average return per trade was **0.57%**.
- **Profit Factor**: The profit factor (ratio of gross profit to gross loss) was **2.2**, indicating that profitable trades outperformed losing trades by a factor of 2.2.
- **Maximum Drawdown**: The maximum drawdown during the testing period was **19.8%**, reflecting the largest peak-to-trough decline in portfolio value.

3.2. Replicated strategy Performance

In this section, we provide an in-depth analysis of the backtesting process for our Stochastic Oscillator trading strategy done using Backtrader framework. We applied these methods to historical data from the S&P 500, using the SPY ETF as a proxy for our analysis.

We used historical price data of the SPY ETF January 1, 1993 - April 08, 2024. The dataset consisted of daily price information, including open, high, low, close prices, adj close and volume.

Source and Data Details

- **Source:** Yahoo Finance
- **Asset**: SPDR S&P 500 ETF Trust ("SPY")
- **Timeframe**: Daily
- **Date Range**: January 1, 1993 - April 08, 2024
- **Backtesting Framework**: Backtrader

Backtest Assumptions

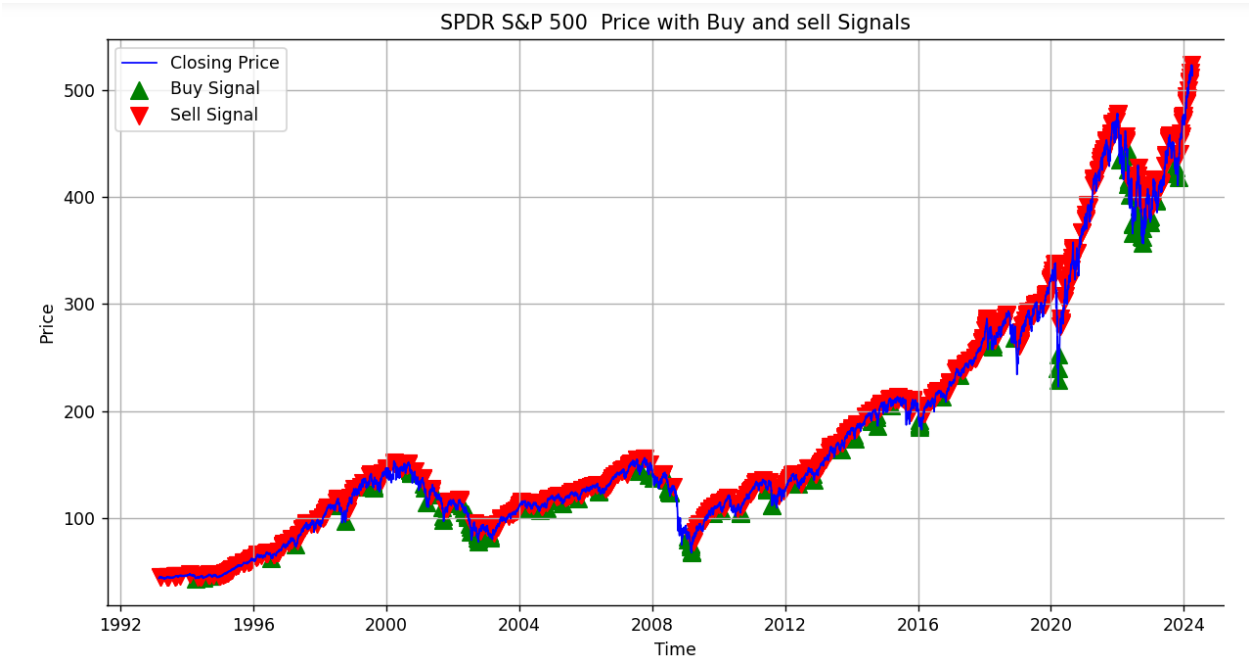
1. **Initial Cash**: The starting portfolio value is set at $100,000.
2. **Transaction Costs**:
    - **Commission**: A percentage of 0.1%. of each trade's value is charged as commission.
3. **Slippage**:
    - **Slippage Percentage**: A percentage of 0.1% is added to account for execution price deviation.

4. **Position Sizing**: The size of each trade is determined based on a 10% of the current portfolio value.
5. The backtest assumed perfect market conditions with no liquidity issues.

Running the Backtest

1. **Data Integration**: Historical price data is integrated into the backtesting framework.
2. **Strategy Execution**: The strategy is executed over the historical data to simulate trading performance.
3. **Performance Metrics**:
    - **Starting Portfolio Value**: The initial value of the portfolio before any trades.
    - **Final Portfolio Value**: The value of the portfolio at the end of the backtest period.
    - **Equity Curve**: The graphical representation of the portfolio value over time.

Price Chart

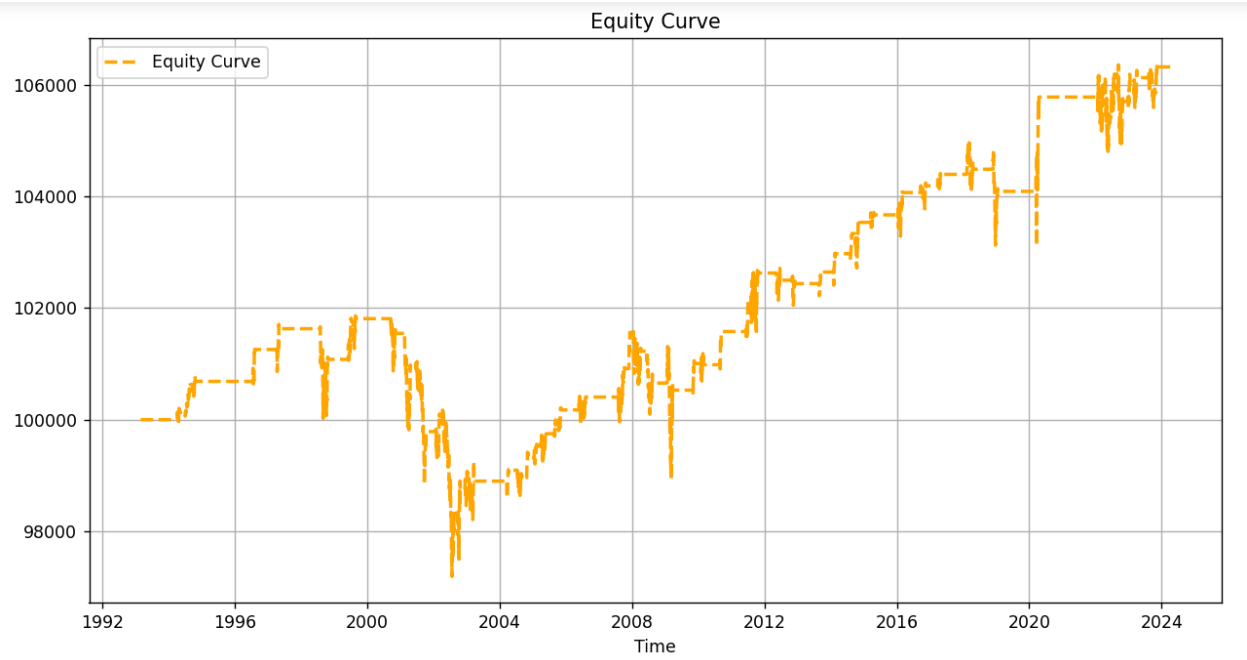


Insight

The chart shows a strong uptrend in SPDR S&P 500 (SPY) from 1992 to 2024, characterized by a consistent increase in price with occasional periods of volatility and consolidation.

Green triangles indicate buy signals, suggesting entry points based on favorable conditions identified by the Stochastic Oscillator. Red triangles mark sells signals, indicating exit points when the strategy detects overbought conditions.

Equity Curve with commission and slippage



Insight

The equity curve shows a general increase in portfolio value from 1993 to 2024. There are ups and downs, with some periods of high growth and others of decline.

The growth rate is uneven, with times of fast gains followed by slow periods. Overall, the curve trends upward, suggesting a positive return in the long run.

The portfolio increased from $100,000.00 to $106,316.41, representing a gain of $6,316.41. This reflects a positive return on investment over the testing period. The increase in portfolio value suggests that the trading strategy was effective in generating profit during this time.

The following table provides a summary of the replicated strategy's performance metrics over the period range:

Metric	Values
Total Return	**6.32%**
Annualized Return	**0.14%**
Max Drawdown	**-4.58%**

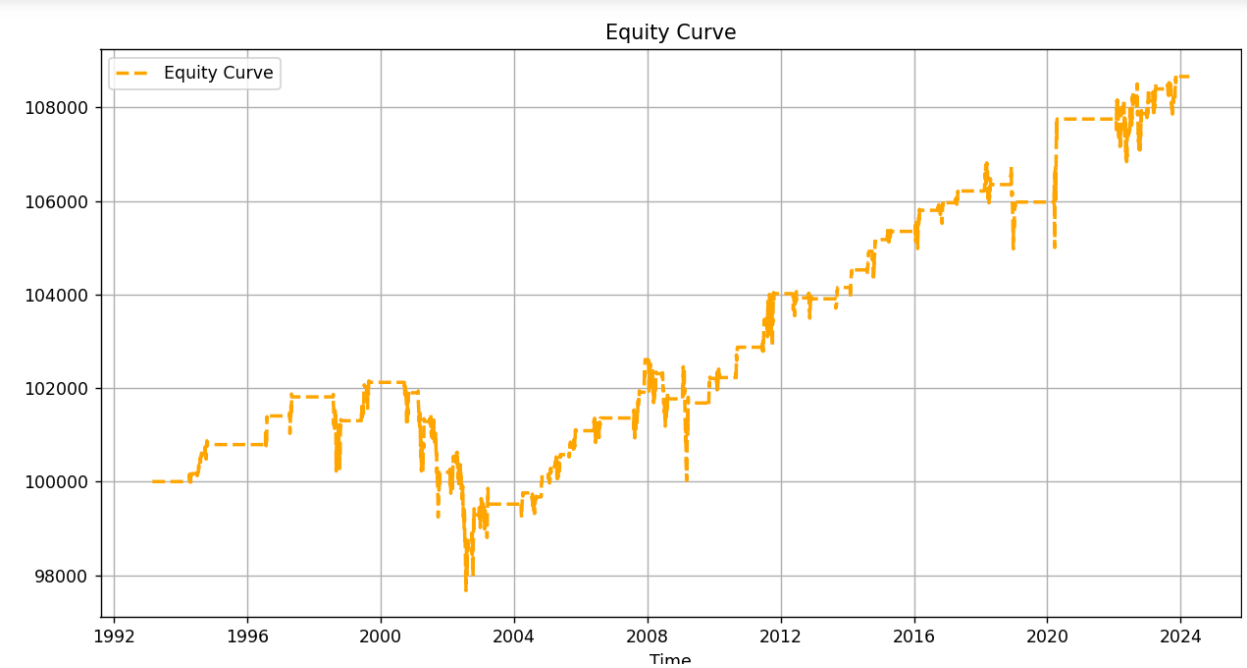
Interpretation

Total Return: 6.32% – The strategy is profitable, but the return is modest given the 31-year backtest.

Annualized Return: 0.14% – Very low, barely outperforming holding cash.

Max Drawdown: -4.58% – Moderate, showing good risk control and loss limitation.

Equity curve without commission and slippage



Insight

This is the equity curve without including commission and slippage. From the graph, we can see there is also a general increase in portfolio value from 1993 to 2024 but importantly, there is a portfolio increase from $106,316.41 to $108661.33 compared the equity curve considering commission and slippage.

The following table provides a summary of the replicated strategy's performance metrics over the period range:

Metric	**Values**
Total Return	**8.66%**
Annualized Return	**0.18%**
Max Drawdown	**-4.39%**

Total Return: 8.66% – The strategy is profitable, but the return is modest given the 31-year backtest.

Annualized Return: 0.18% – Very low, barely outperforming holding cash.

Max Drawdown: -4.39% – Moderate, showing good risk control and loss limitation.

3.3. Optimization

To enhance the performance of the initial Stochastic Oscillator strategy, we conducted a hyperparameter optimization. The goal was to fine-tune the parameters for better alignment with market conditions and improve overall returns [3]. The following parameters were selected for optimization:

- **Oversold Threshold**: Tested values ranging from 10 to 30.
- **Overbought Threshold**: Tested values ranging from 70 to 90.
- **Stop-loss**: Considered values of 2%, 5%, and 10% to limit downside risk.
- **Take-profit**: Tested profit levels of 5%, 10%, and 20% to capture gains.
- **Position Size**: Allocated 5%, 10%, and 20% of the portfolio per trade to evaluate risk and reward balance.

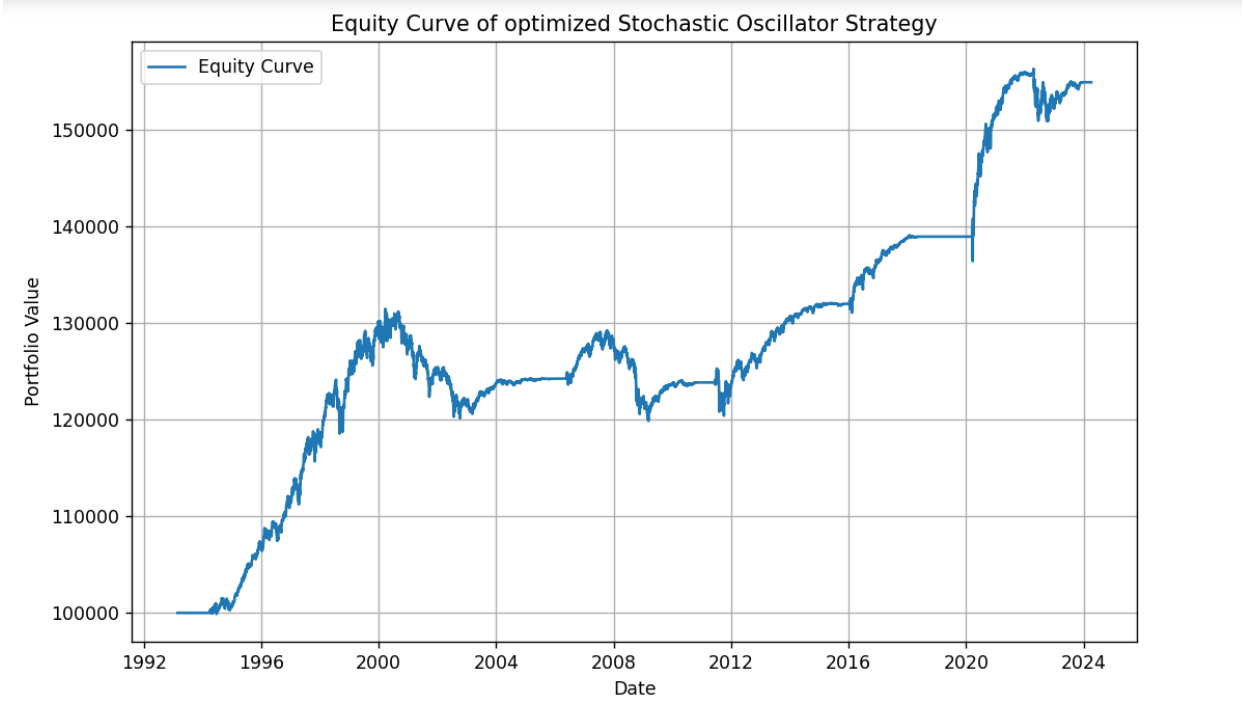
We employed Backtrader's **optstrategy** functionality to perform the optimization, which tested all possible combinations of the selected parameters [4]. Each combination was backtested using historical data for SPY from 1993 to 2024.

Best Parameter Set

After the optimization process, the best-performing combination of parameters was:

- **Oversold Threshold**: 15
- **Overbought Threshold**: 70
- **Stop-loss**: 10%
- **Take-profit**: 10%
- **Position Size**: 20% of the portfolio

Equity Curve



Insight

The optimized Stochastic Oscillator strategy applied to the S&P 500 (1992–2024) shows positive long-term growth, with an equity curve that increased from $100,000 to over $150,000 by 2024. This reflects a significant return on investment over the 30+ year period.

The strategy effectively captures key market events, including the dot-com bubble (2000-2002), the 2008 financial crisis, and the COVID-19 crash and recovery in 2020. Additionally, it has shown strong performance since 2016, with faster growth during this period.

The following table provides a summary of the replicated strategy's performance metrics over the period range:

Metric	**Values**
Total Return	**54.93%**
Max Drawdown	**-8.82%**
Win Rate	**45.92%**
Sharpe Ratio	**-0.21**

Total Return (54.93%): The strategy increased the portfolio by 54.93% over the entire backtest period.

Annualized Return (0.98%): The average yearly growth was 0.98%, indicating slow, steady growth.

Annualized Volatility (2.61%): The portfolio had low fluctuations, meaning it was relatively stable.

Sharpe Ratio (-0.21): The strategy underperformed on a risk-adjusted basis, suggesting it wasn't worth the risk.

Max Drawdown (-8.82%): The portfolio lost 8.82% at its worst point, which is moderate.

Win Rate (45.92%): The strategy won 45.92% of the time, losing slightly more often but still achieving overall profitability.

4. Discussion

4.1. Pre-optimization Strategy Performance Comparison

The two-trading strategy reflect contrasting investment performances over a similar time period by referring to the equity curves.

The publisher's portfolio shows a much stronger performance. The portfolio grows from $100,000 to over $2 million, a huge increase of around 1900%. The growth is steady, with smaller dips and quicker recoveries. The portfolio performs better in tough market times and grows rapidly, especially in the later years. This suggests a more stable and successful strategy.

Our strategy's portfolio starts with $100,000 and grows to just $106316.41 over 32 years, giving a small return of around 6%. There are big drops during events like the dot-com bubble (2000-2002) and the 2008 financial crisis. The curve also shows more ups and downs in recent years, especially around 2020 during the COVID-19 pandemic. This shows a strategy that reacts more to the market and grows slowly with a lot of volatility. Since this is an initial back testing, optimization is a crucial aspect that can increase performance.

4.2. Post-optimization Strategy Performance Comparison

After optimizing the Stochastic Oscillator strategy, the performance saw a notable improvement compared to the initial backtest. The total return increased significantly from **6.32% to 54.93%**, and the annualized return rose from **0.14% to 0.98%**, indicating that the optimized parameters allowed the strategy to better capture profitable opportunities over the same period.

However, this optimized performance still falls short when compared to the publisher's strategy. Our strategy's results differ from those in the article due to several key factors. First, the article uses specific trading rules, particularly regarding entry and exit conditions based on %K/%D crossovers, which may differ from the parameters we optimized, leading to variations in performance. Additionally, the lookback periods and thresholds for overbought and oversold levels in our backtest may not align with the optimal settings used in the article, affecting the results. Another important factor is the execution tools: while we used Backtrader with yFinance data, the article utilized Amibroker, which might handle data and trade execution differently, contributing to discrepancies.

4.3. Lessons Learned

Through the process of developing and optimizing the Stochastic Oscillator strategy, several valuable lessons were learned. First, it became clear that **strategy parameters** such as lookback periods, overbought/oversold thresholds, and stop-loss settings have a significant impact on overall performance. Fine-tuning these parameters can lead to noticeable improvements in profitability and risk management. Additionally, the **importance of risk management** was reinforced, as implementing stop-losses and take-profit levels proved crucial in minimizing large drawdowns and protecting capital.

5. Conclusion

The optimization of the Stochastic Oscillator strategy led to noticeable improvements in total return, annualized return, and overall portfolio performance. Despite these enhancements, the strategy still underperformed when compared to the results discussed in the article. Factors such as differences in trading rules, parameter settings, execution platforms, and data quality played a significant role in these discrepancies.

Moving forward, further refinements to the strategy such as incorporating additional filters or indicators and testing in different market environments could yield better results. Overall, the optimization process provided valuable insights into the strengths and weaknesses of the Stochastic Oscillator strategy, laying the foundation for future improvements.

7. Biography

[1] O. Groette, "Quantified Strategies," 08 04 2024. [Online]. Available: https://www.quantifiedstrategies.com/stochastic-indicator-strategy/.

[2] M. George C. Lane, Stochastics: A Useful Tool in the Toolbox. Technical Analysis of Stocks and Commodities Magazine., G.C, 1984.

[3] U. Jaekle and Tomasini, "Variation of the Input Parameters: Optimisation and Stability Diagrams," in *Trading Systems: A New Approach to System Development and Portfolio Optimisation*, 2019.

[4] "ChatGPT," OpenAI, September 2024. [Online]. Available: https://chat.openai.com/.